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Magnetostratigraphy of Eocene–Oligocene boundary sections in Italy: No evidence for short subchrons within chrons 12R and 13R

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Abstract

The marine magnetic polarity record shows many small-scale anomalies that could be due to short normal polarity subchrons within chrons 12R and 13R. New examination of published magnetostratigraphic results from a section near Gubbio shows that suspected short chrons in this section are probably not real. The Eocene–Oligocene boundary in the stratotype section at Massignano lies just above a tentatively identified normal magnetozone that might correlate with a possible short normal subchron within chron 13R. In a new investigation of the weathered marlstones in this section, acquisition and thermal demagnetization of IRM reveal a magnetic mineralogy dominated by magnetite, with minor amounts of maghemite and hematite. Vector analysis of the natural remanent magnetization discloses a complex structure in some samples: at least two and up to four components were isolated. The magnetostratigraphy based on a magnetite component correlates well to the main features of the chron 13N–16N polarity sequence. No short polarity subchrons are found, although the resolution is inadequate to exclude the possibility that they exist. In particular, we did not find a short normal polarity magnetozone previously located close to the Eocene–Oligocene boundary. The results of this study favour the interpretation that low-amplitude magnetic anomalies represent geomagnetic intensity fluctuations rather than unresolved short polarity intervals.

1. Introduction

The Late Cretaceous and Cenozoic record of geomagnetic polarity is lacking in short polarity intervals [1]. It is not yet known if the short polarity chrons are absent because they never occurred or because, lasting less than 20 k.y., they are too short to be resolved in the marine magnetic anomaly sequence, which serves as the best available record of geomagnetic polarity for the past 83 m.y. Magnetostratigraphic investigations in continuously deposited sediments have not resolved the issue. Detailed investigations of inclination records in cores from DSDP Site 522 showed no short polarity subchrons within chron 12R where eight were expected, but three subchrons were found in chron 13R [2].

A thorough revision of the marine magnetic database [3], including new evaluation of the relative lengths of polarity chrons and subchrons, has given a definitive polarity sequence and a new magnetic polarity timescale (hereafter referred to as CK92). Cande and Kent [4] have addressed the origin of small-scale magnetic anomalies ('tiny

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wiggles') in this sequence. They concluded that the tiny wiggles are more likely due to geomagnetic intensity fluctuations than to short polarity chrons. A recent magnetostratigraphic study of deep-sea sediments from DSDP Site 522 supports the intensity-fluctuation hypothesis [5]. The CK92 timescale indicates the possible existence of many short polarity subchrons ('cryptochrons') in the early Oligocene. Eight cryptochrons lie in chron 12R, one in chron 13N and four in chron 13R; one of the latter lies very close to the Eocene–Oligocene (E/O) boundary.

The E/O boundary was used as one of the time calibration points for the new CK92 timescale. An interpolated age of 33.7 ± 0.5 m.y. was obtained from K/Ar and Rb/Sr dating of biotites from ash layers in magnetostratigraphic sections in Umbria, in central Italy [6]. Several short magnetozones were reported in these sections, which are located in the Contessa gorge, near Gubbio in the region of Umbria, and in a small quarry at Massignano, near Ancona in the region of Marche (Fig. 1). On the basis of its lithological and paleontological completeness the Massignano section was proposed [7,8] and selected as the stratotype section for the E/O boundary. In this paper we critically evaluate the published data from these sections and present new magnetostratigraphic results from the Massignano section.

2. Previous results

In a combined magnetostratigraphic and biostratigraphic study in an active quarry in the Contessa gorge the E/O boundary was found to lie in the lower part of the *Scaglia cinerea* formation, just above the gradual transition from the underlying *Scaglia variegata* formation. The magnetostratigraphy of this section placed the E/O boundary within reversed polarity chron 13R [9]. This result was confirmed in later magnetostratigraphic investigations elsewhere in the Contessa gorge—in a section referred to as Contessa Highway III [10]. Three short normal polarity subchrons within chron 13R and one in chron 12R were reported in this section. However, the published declination and inclination data [10] show several aberrations (Fig. 2) and do not agree in all points with the interpreted polarity sequence, which is based on the computed latitude of the virtual geomagnetic pole (VGP) during deposition.

The use of VGP latitude allows the combination of declinations and inclinations in a single index of polarity. The technique assumes that the observed mean direction in a section corresponds to a geocentric axial dipole field. This requires correcting the observations for possible tectonic rotation by referring the declinations to the declination of the mean direction. Subsequent plate motion is taken into account by using the paleolatitude of the site (calculated from the inclination of the mean direction) instead of its present latitude. These corrections may have been omitted in computing the profile of VGP latitude for the Contessa Highway III section. A revised plot of VGP latitude derived from the published data shows no solid evidence for short chrons in 13R and 12R (Fig. 2), although it indicates a short reversed subchron in chron 13N, which agrees

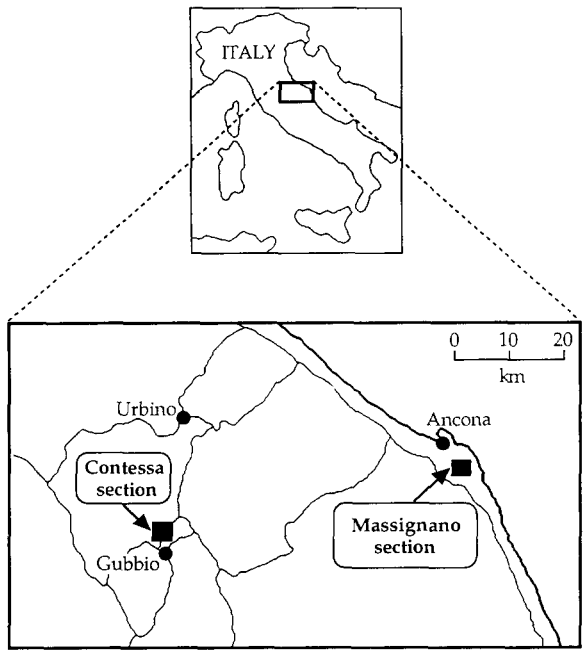


Fig. 1. Sketch map showing the locations of the Massignano and Contessa sections in north-central Italy.

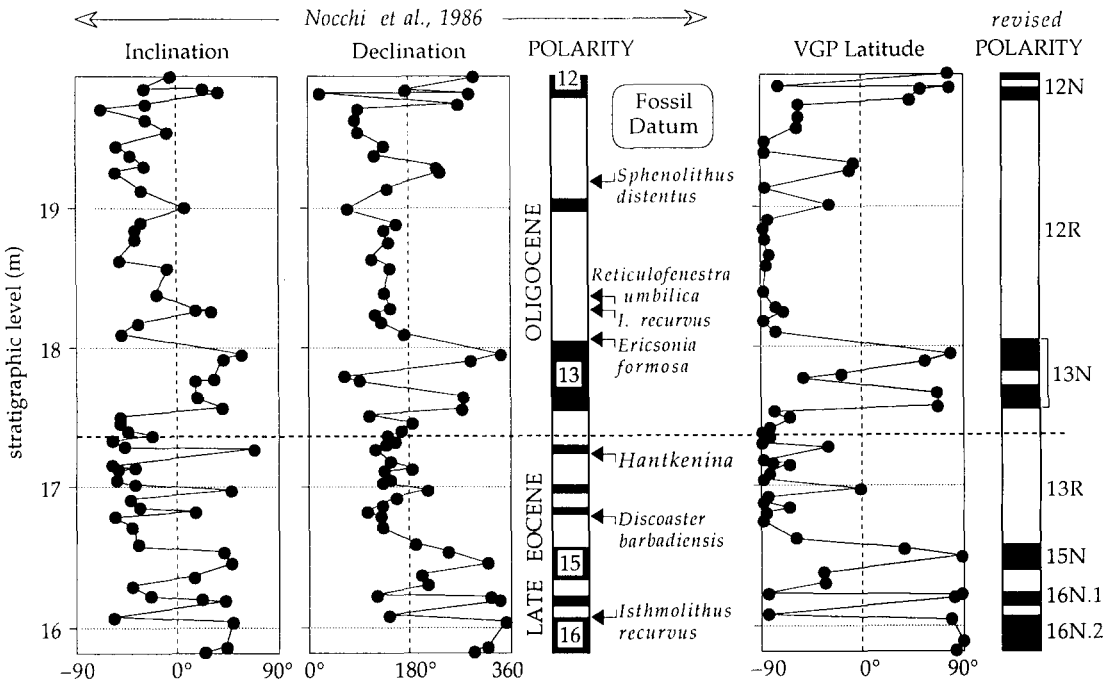


Fig. 2. Inclinations and declinations of magnetostratigraphic samples, and fossil datum levels (calcareous nannofossils, except for the planktic foraminifera *Hantkenina*) from the Contessa Highway III section (after [10]). The VGP latitudes and polarity zonation were recomputed for this study.

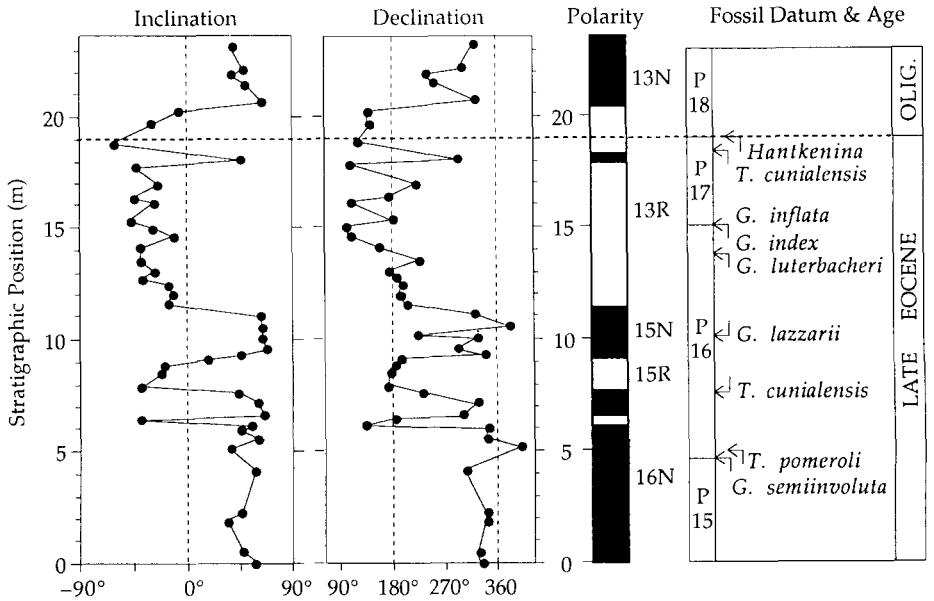


Fig. 3. Inclinations, declinations and polarity zonation of magnetostratigraphic samples, and datum levels of planktic foraminifera in the Eocene–Oligocene boundary stratotype section at Massignano (after [12]).

with a suggested cryptochron in CK92. It should be noted, however, that this reverse chron is determined entirely by aberrant declinations; the inclinations show no discrepancy and so the event is not well supported.

Following detailed biostratigraphic studies in sections in the Marche–Umbrian basins the extinction of the planktic foraminifera *Hantkeninidae* has been suggested as the most reliable event for recognizing the E/O boundary [11]. In the measured Massignano stratotype section the E/O boundary is located at the 19 m level. Ash layers at 7.2, 12.9 and 14.7 m gave radiometric ages of 35.5 ± 0.3 m.y., 34.3 ± 0.2 m.y. and 34.6 ± 0.3 m.y. respectively. These three dates formed part of the determination of an optimum age of 33.7 ± 0.5 m.y. for the E/O boundary [6]. The correlation of the boundary to the polarity sequence was used as a calibration point for the CK92 polarity timescale [3]. In the magnetostratigraphy of the Massignano stratotype section (Fig. 3) a single sample with normal polarity at the 18 m stratigraphic level, just below the E/O boundary, was interpreted as a short normal polarity subchron within chron 13R [12]. If real, this subchron could constitute a convenient marker for the boundary and would have implications for the controversial origin of low-amplitude marine magnetic anomalies.

3. Lithology and sampling

In the quarry in the Contessa gorge, which is still being actively exploited, the *Scaglia cinerea* is a dark grey marlstone. In a gradual transition, it replaces the *Scaglia variegata*, which is a variously hued, marly limestone containing reddish tinted beds. The magnetic properties of the *Scaglia cinerea* in this section are largely due to magnetite and are straightforward [9]: alternating field and thermal demagnetization readily define the characteristic remanent magnetization (ChRM). In contrast, the rocks in the small abandoned Massignano quarry have a yellowish hue due to weathering. Under this superficial discolouration the *Scaglia cinerea* samples were greenish brown rather than dark grey.

The magnetic properties of samples from the section have been partially documented. Montanari and Bice [13] found several types of behaviour during alternating field and thermal demagnetization of the natural remanence. The study relied heavily on principal component analysis of individual directions and was handicapped by weakness and instability of the remanence, overprinting and viscous components. The rock magnetic properties were not investigated further.

The bedding in the Massignano quarry dips gently and uniformly at 23° toward the southwest with an azimuth of 227°. The rocks are fissile on bedding partings, which made sampling difficult. The vagaries of the outcrop limited the continuity and interval of the sampling. Oriented hand samples were taken at somewhat irregular intervals. Because it has been proposed as a stratotype section for the E/O boundary, the metre levels of the measured section in the quarry have been marked by metal plaques. This enabled us to correlate our sampling with the previous investigations. Extra samples were taken at and adjacent to the 18 m level where a short normal magnetozone had been reported. Cylindrical samples were later prepared by drilling 2.54 cm diameter cores along the down-dip direction and cutting them to the standard height of 2.25 cm. A total of 32 levels at an average separation of 70 cm was sampled in the quarry; sample spacing between 17 and 20 m, near the E/O boundary, averaged 35 cm.

4. Rock magnetism

Coercivity and thermal unblocking characteristics [14] were used to investigate the magnetic mineralogy of the Massignano samples. The stepwise acquisition of isothermal remanent magnetization (IRM) was observed in progressively increasing fields up to 1 T. At this stage a second IRM in 0.2 T was induced at right angles to the first to remagnetize the soft fraction; in a few samples three orthogonal IRM components were induced in, successively, 1, 0.65 and 0.2 T. The composite IRM was then demagnetized thermally

and the unblocking of the soft (coercivity < 0.2 T), intermediate (0.2–0.6 T) (if used) and hard (0.6–1 T) fractions was plotted against temperature (Fig. 4). This procedure was applied to each stratigraphic level sampled in the Massignano quarry. None of the samples reached saturation

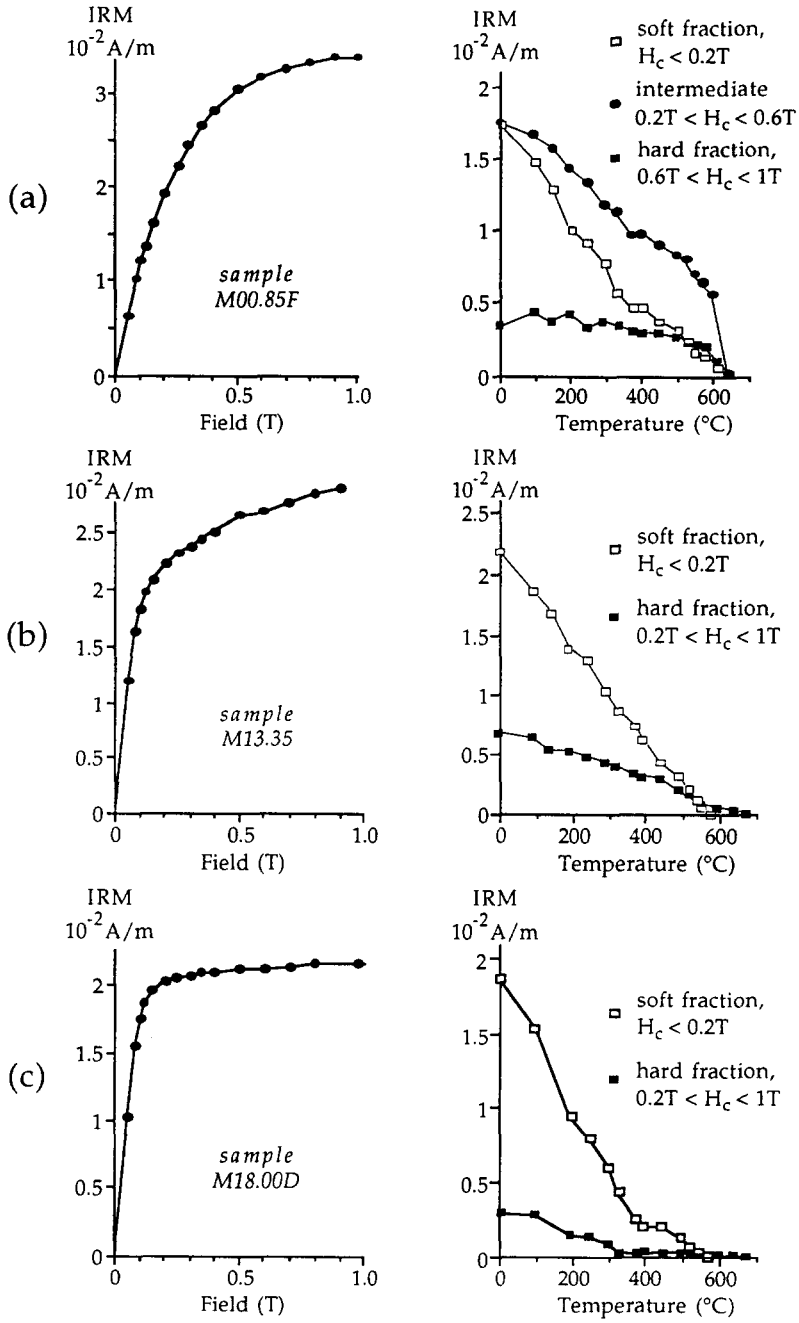


Fig. 4. Progressive acquisition of isothermal remanent magnetization (IRM) and thermal demagnetization of composite IRM in (a) *Scaglia variegata* and (b and c) *Scaglia cinerea* samples.

in 1 T although some were nearly saturated in 0.2 T.

Samples of the *Scaglia variegata* formation at the base of the section (Fig. 4a) contain high-coercivity (hard and intermediate) fractions that are only removed thermally at 630°C and are probably due to hematite. The low-coercivity (soft) fraction is dominated by a phase that is mostly demagnetized below 600°C and is probably magnetite; part of the soft fraction persisted to temperatures above 600°C and may represent a low-coercivity (possibly multidomain) phase of hematite.

Most samples of *Scaglia cinerea* showed a clear distinction between a soft fraction that unblocked at 580°C and is identified as magnetite and a hard fraction that unblocked at about 660°C and is identified as hematite (Fig. 4b). The IRM acquisition of the sample at 18 m (Fig. 4c) shows mainly low coercivity, with only a very small fraction harder than 0.2 T. As well as magnetite, a phase is present that demagnetizes below 400°C, and this may be maghemite. This phase was suspected in several other samples from the section. The possible maghemite and hematite phases are much less significant than the dominant presence of magnetite.

5. Characteristic remanent magnetization

At each stratigraphic level one sample was demagnetized progressively in alternating magnetic fields up to 85 mT using a Schonstedt alternating field (AF) demagnetizer; another sample from the same level was demagnetized thermally. The components of the natural remanent magnetization (NRM) were analyzed using vector diagrams [15].

During AF demagnetization two or three components were found in most samples (Fig. 5). Usually a very soft component was removed in a low peak field of less than 2 mT; the vector demagnetization trajectory was curved up to 16–20 mT, above which a linear path to the origin of the plot was obtained up to 85 mT (Fig. 5, left). Some samples showed linear behaviour from 2 to 20 mT; a curved trajectory was then found to 55 mT, and a linear path up to 85 mT (Fig. 5, right). The NRM of these samples evidently contained

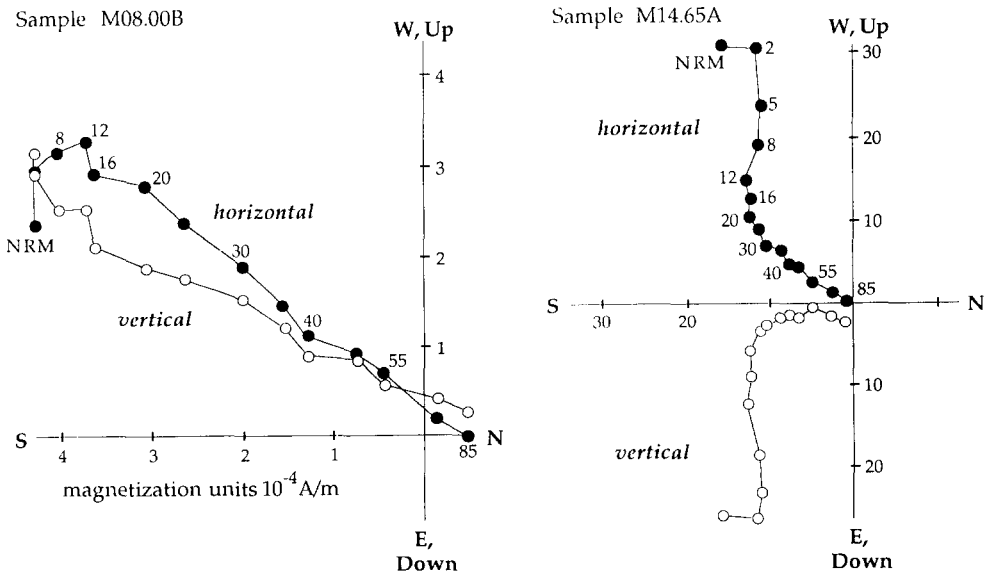


Fig. 5. Vector diagram showing alternating field demagnetization behaviour of *Scaglia cinerea* samples. ● = Horizontal plane; ○ = east–west vertical plane.

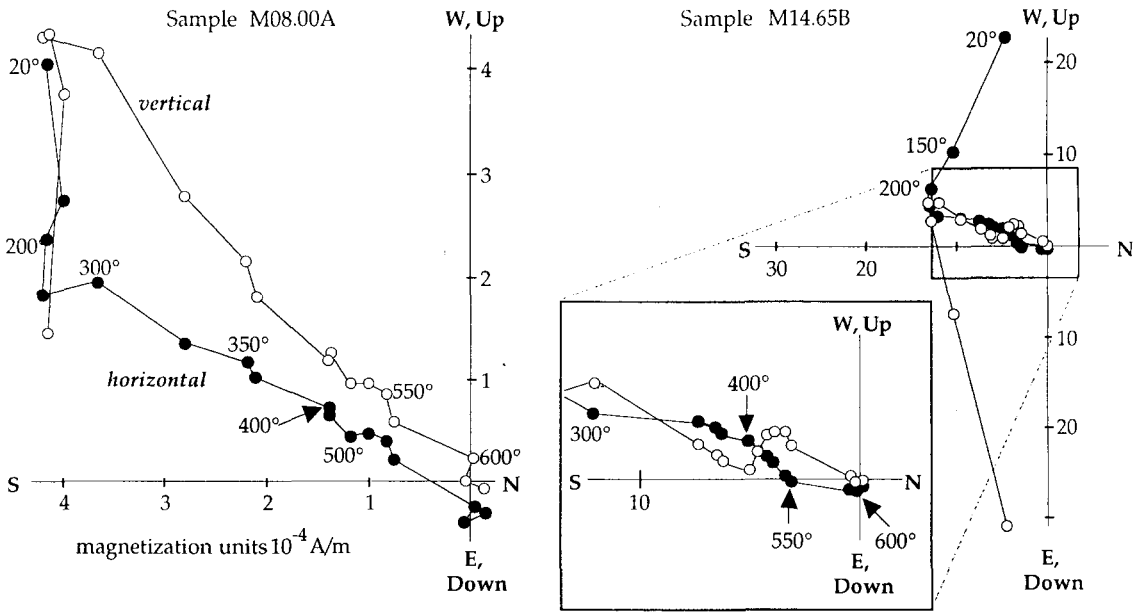


Fig. 6. Vector diagram showing thermal demagnetization behaviour of *Scaglia cinerea* samples. Symbols as in Fig. 5.

components with overlapping coercivity spectra that could not be resolved optimally by AF demagnetization. The straight line defining the final component generally passed by the origin of the vector diagram, indicating the presence of an unresolved high-coercivity component.

Thermal demagnetization, using a Schonstedt shielded oven, revealed only two significant components of NRM in many samples (Fig. 6, left). A low-unblocking temperature component with a direction close to the present field was removed below 200–250°C, and a component with a wide

Table 1
Summary of vector mean directions of magnetization components. k = dispersion parameter; α_{95} = radius of 95% confidence about the mean [16]

NRM		Geographic Coordinates				Bedding Corrected			
Component	N	Dec	Inc	α_{95}	k	Dec	Inc	α_{95}	k
A (20-250°C)	32	321	68	7.1	13.82	276	60	7.2	13.31
C (400-550°C)	10	298	71	11.9	17.41	261	58	12.0	17.04
B (300-400°C)	19	20	28	10.3	11.51	10	47	10.6	11.08
D (>550°C)	6	20	17	7.7	76.16	15	35	6.7	100.63
W (200-600°C)	14	23	21	7.7	27.76	17	40	7.5	28.91
ChRM	29	21	26	7.3	14.24	13	45	7.4	14.02

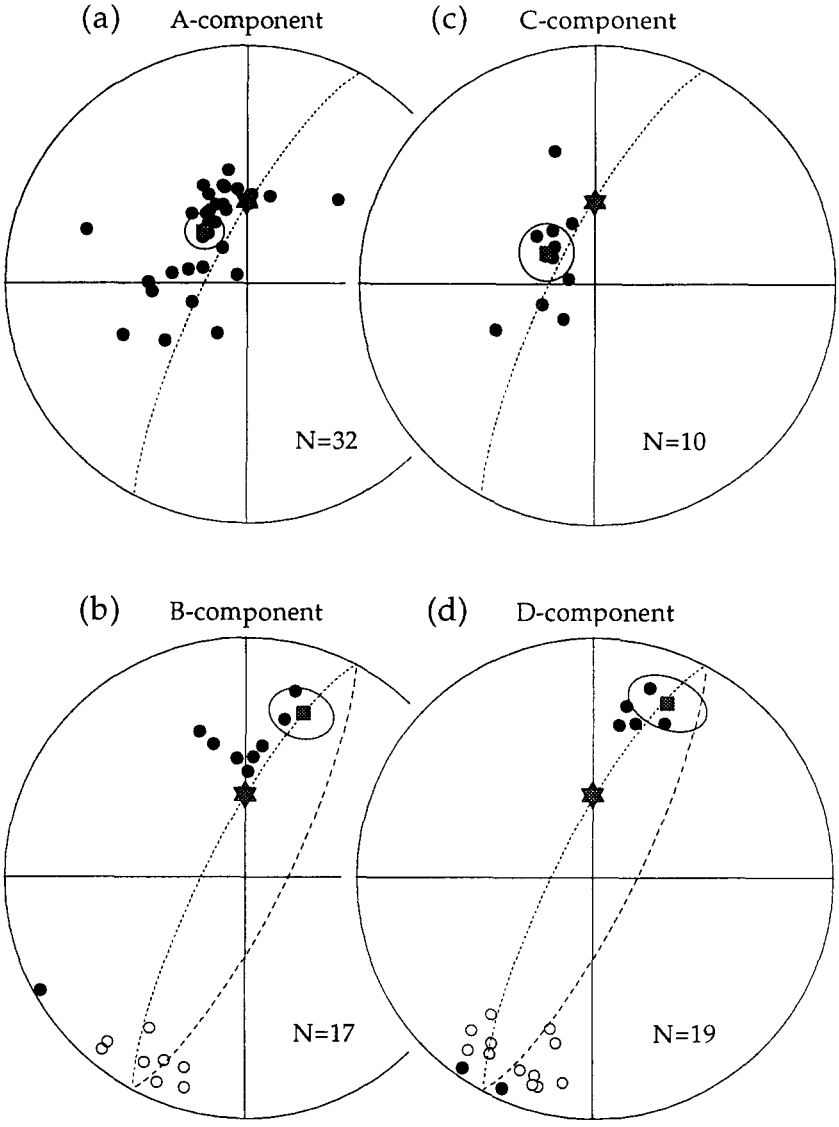


Fig. 7. Equal-area stereograms of components of remanent magnetization in paleomagnetic samples from the Massignano quarry. Solid points and dotted great circle are on the lower hemisphere; open points and dashed great circle are on the upper hemisphere; the star is the present axial dipole field direction; squares are the mean directions of the components.

range of unblocking temperatures was resolved from about 250°C to the highest temperature used; we refer to this as the W-component. About one third of the samples exhibited a more complex NRM structure (Fig. 6, right) with four components. As in other samples, an initial component (A-component) was removed below 200°C. Other components were resolved from 250–300°C to 400°C (B-component), from 400°C to 500–550°C (C-component), and above 550°C (D-component).

The directions of these components were plotted on stereographic projections (Fig. 7). The statistical parameters of the distributions (Table 1) were computed with standard procedures [16]. As the W-component also persisted to high tem-

peratures, its directions are plotted as D-components. The B- and D-components showed both normal and reverse polarities. These formed almost antipodal groups for the D-component (Fig. 7d). However, they were not antipodal for the B-component (Fig. 7b); the B-reverse groups agreed well with the D-reverse group, but the B-normal group lies between the B-mean and the present axial dipole field (ADF) direction. The mean directions of the B- and D-components did not differ significantly (Table 1). For the characteristic remanent magnetization (ChRM) we chose preferentially the W-component with a wide unblocking temperature range, or the high unblocking temperature D-component; in samples where these components were not readily defined we used the B-component.

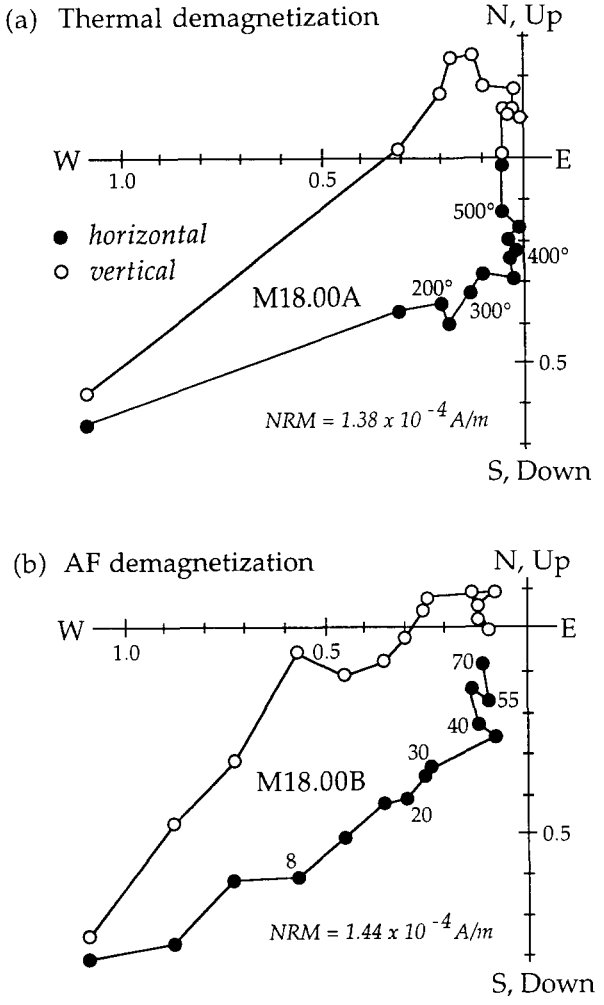


Fig. 8. Vector diagrams of (a) thermal and (b) alternating field demagnetization of sister samples from the 18 m level where a normal polarity magnetozone had been postulated. Symbols as in Fig. 5.

For rocks of Oligocene age at this locality a recent African apparent polar wander reference path [17] predicts a declination of 6° and an inclination of 56°. The mean direction of ChRM in the Massignano quarry (Table 1) is appreciably different from the expected value. The inclination of 45° is virtually the same as found in Paleogene carbonates in the Contessa gorge [9], but is 11° too shallow. The declination of 13° is rotated 7° clockwise, in contrast to Paleogene and Late Mesozoic directions from the Marche–Umbria region of central Italy, which usually show anticlockwise rotations [18].

The A- and C-component directions are dispersed along a great circle that passes through the B-, D- and ChRM mean directions and the present ADF direction. Many of the A-component directions are close to the ADF direction; the remainder, together with the C-component directions, are streaked along this great circle between the ADF direction and the reverse ChRM direction (Figs. 7a and c). We conclude that the A- and C-components are normal overprints, primarily due to partial remagnetization in a recent field, and contaminated by an unremoved element of the B-component. The A-component is probably a simple viscous overprint. The C-component is defined in a range of unblocking temperatures above the Curie point of maghemite. It was not possible to identify the magnetic mineral carrying this component definitively. Its blocking temperature spectrum was narrow, suggesting it may be carried by a limited range of grain sizes. The predominance of magnetite in these rocks suggests that it might be a secondary phase of magnetite. Examples have been reported where secondary magnetite formed in carbonate rocks from iron-rich fluids that were able to permeate the rocks during tectonic deformation [19–21]. Thus, the C-component may be associated with a secondary magnetite phase that

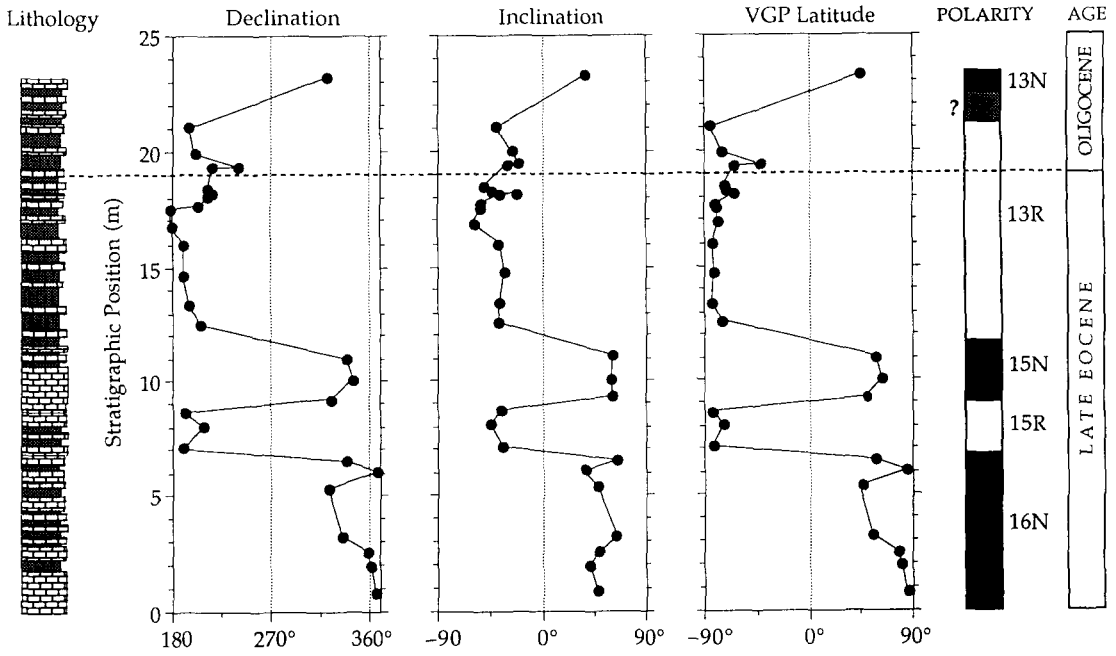


Fig. 9. Magnetostratigraphy of the new investigation of the Massignano section.

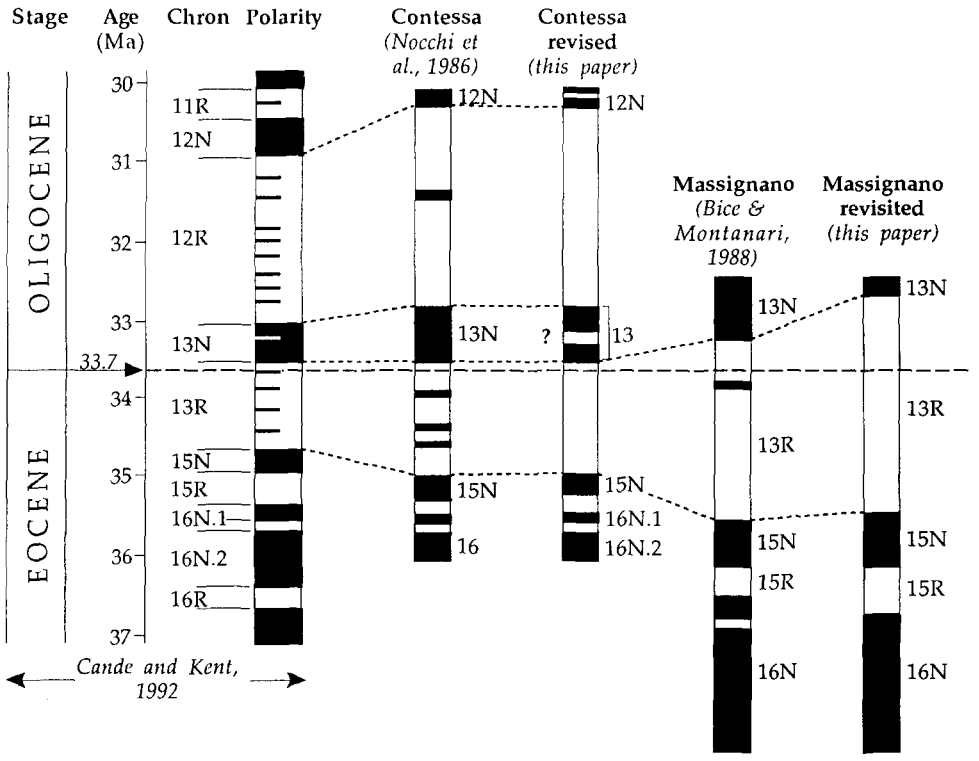


Fig. 10. Summary of Eocene–Oligocene boundary magnetostratigraphies.

originated during deformation in the late Tertiary. If so, this is a chemical remanent magnetization (CRM). Because CRM is a more efficient magnetization process than the post-depositional remanence (pDRM) believed to carry primary remanence in these rocks, only a small secondary phase might be enough to compete with the fraction of pDRM in the 400–500°C temperature range. The direction of the C-component may then be a composite of the original pDRM and the remagnetized CRM superimposed on it.

Because of the potential importance of samples from the 18 m level at which a short normal polarity chron was postulated [12], the AF and thermal demagnetization characteristics of this level are shown in Fig. 8. Each type of treatment resolves two components: an initial component magnetized downward to the west-southwest (equivalent to the overprinted C-component), and a more stable reverse component magnetized upward to the south. The ChRM of this level has, therefore, reverse polarity like the rest of chron 13R in which it lies.

6. Discussion

The stratigraphic plots of declination, inclination and VGP latitude (Fig. 9) show a clear pattern of magnetozones that can be easily identified by comparison with the marine record as chrons 13N–16N. The boundary between 13N and 13R is not well defined because only one sample (at 22 m) could be taken between 21 and 23.2 m due to the poor condition of the outcrop. Unfortunately, this sample had unstable magnetic properties that prohibited determination of a stable direction. The thickness of chron 13R may be as much as 12 m, or as little as 10 m, in the Massignano section. This means that the E/O boundary at 19 m could fall as much as 30%, or as little as 20%, from the top of chron 13R. This is a minor uncertainty, with no consequences for the calibration of the CK92 timescale (in which the boundary was correlated at 14% from the top of 13R).

Our reappraisal of the magnetostratigraphic data from the Contessa Highway III section and the original and new sets of data from the Massignano stratotype section (Fig. 10) generally agree well with the polarity sequence in the CK92 timescale. Our revision of the Contessa Highway III magnetostratigraphy of Nocchi et al. [10] shows two reversed samples within chron 13N, which might correspond to a cryptochron in CK92. However, the polarity of these samples is determined entirely by their declinations. The inclinations show no discrepancy; hence misorientation of the samples might be suspected. Thus, the reality of this 'event' is dubious. We conclude that the magnetostratigraphies of the Contessa and Massignano sections show no convincing evidence of short polarity chrons.

A weakness common to each of these magnetostratigraphic investigations in the *Scaglia cinerea* is the comparatively wide sampling interval. This is because sampling of the friable marls was restricted to beds that did not disintegrate. This resulted in an irregular and unfortunately wide sampling interval. Closer sampling should be possible in the fresher material in a continuous core recently drilled through the *Scaglia cinerea* at Massignano [22]. Magnetostratigraphic analysis of this core has recently started.

The durations of the cryptochrons in CK92 were estimated to be about 8–17 k.y. At Massignano chron 13R, which according to CK92 lasted 1.1 m.y., is represented by a magnetozone that is 10–12 m thick. The sedimentation rate of about 1 cm/k.y. means that a sample spacing of 5–10 cm would be needed to detect cryptochrons. In our study the mean sampling interval was 70 cm, whereas in Bice and Montanari [12] it was 50 cm. Except at the 18 and 19 m levels, our samples did not come from the same stratigraphic levels as those of Bice and Montanari [12]. Combining the two data sets gives a mean sampling interval in the Massignano quarry of 30 cm, equivalent to about 30 k.y. This is still too wide to detect all four possible cryptochrons within chron 13R, or the eight cryptochrons within 12R. However, if all the cryptochrons are real, it is surprising that not one of them was detected in these sections. In the absence of evidence for the existence of short polarity chrons, we favour the interpretation [4,5] that the tiny wiggles in the marine magnetic

record are expressions of geomagnetic intensity fluctuations.
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